# DMM6500 behaviour the manuals do not state

Keithley publishes an example repository at `github.com/tektronix/keithley`. These notes come from reading it — `Instrument_Examples/`, `Application_Specific/`, `TTI_Apps/` and `Drivers/`, 217 TSP files and their host-side callers — and checking every call it makes against the DMM6500 reference manuals (`DMM6500-901-01A` April 2018 and `DMM6500-901-01B` September 2019), the user manual, and the display and TSP-app documents that ship beside the examples.

A fact earns a place here only if an example demonstrates or depends on it **and** the reference manual is silent, lists the call without explaining it, or contradicts it. Where a vendor display document is the contradicting source rather than the reference manual, this file names it.

The instrument is the arbiter, and on the trigger model it has to be. Keithley's own application code never reads a `trigger.` setting back — `GAS/MS01_SetupAndRun.tsp` performs about twenty writes under `trigger.` and `tsplink.` and checks none of them, no `getblocklist()`, no attribute read, no event-log drain. A refused write under `trigger.` files an event rather than failing, so an example shows what Keithley intended, not what the firmware took. That is why every probe below reads a value back.

**Every claim here has been to the instrument**, so nothing is left standing as a hypothesis. What is not measured is refuted, or out of reach on this bench for a stated physical reason rather than for want of trying. A few claims are measured in part and say so in their own sentence.

**There is deliberately no tally in this paragraph.** Two writers re-derived one by hand within an hour and it went adrift twice, once by a claim and once by two statuses. Count the status tokens if you want the figures; a number nobody can recompute from the file is worse than no number.

| Status | Meaning |
|---|---|
| `MEASURED` | Confirmed on this DMM6500, firmware 1.7.17a. |
| `REFUTED` | The instrument disagreed. The claim stays, with what actually happens. |
| `UNTESTABLE HERE` | Out of reach on this bench, with the reason in the same sentence — a human at the front panel, a second instrument, an accessory card, a power cycle. Never "hard". |

Three claims were **cut** rather than given a status, because the manual turns out to state each in plain English and they failed this file's own bar: digitize having no autorange, stated at `DMM6500-901-01B:7730` and again at 8402 with `dmm.digitize.range` 's own Details calling the range "fixed"; `buffer.save()` taking the `SAVE_*` constants, whose own Example section uses them twice; and `localnode.showevents` emptying the event log as it echoes, measured as ten 4915 provoked under `showevents = 1` leaving `eventlog.getcount(eventlog.SEV_ALL)` at **0** and `eventlog.next` returning nothing, where the same provocation at 0 leaves all ten readable — which `eventlog.next()` 's own Details state: "When `localnode.showevents` is enabled, this command never returns an event because those events are automatically read and sent to the remote interface", and `eventlog.getcount` counts the **unread**. So any event count taken while the echo is on reads zero, and the manual says so. All three were true, and none was undocumented.

Under each claim is its bench probe, indented and in backticks. Four rules hold across them. A probe that changes instrument state carries its own restore step. No probe rests on a `pcall` verdict — each one reads the attribute back, reads `getblocklist()`, or drains and prints `eventlog.next(eventlog.SEV_ALL)`; on the file API a `pcall` verdict is a constant, because no `file.*` call raises, and under `trigger.` a refused `setblock` and a refused `buffer.make` both return success. Every probe prints one **tagged** value, because the instrument volunteers event lines on the control socket whenever `localnode.showevents` is set, and an untagged read then returns an event where the reply should be and leaves every later read one behind — measured, eight replies adrift. So the probes run with `showevents = 0` and restore it afterwards. A model is polled from **inside** the instrument in every claim but one, because a host query aborts an initiated model; the exception is the claim where that arriving command is itself the experiment. Where a claim needs a front-panel key, the press goes through the instrument's own web panel rather than a finger, and the claim says so.

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## TSP scripting, apps and the host interface

### No DMM6500 manual mentions the display object API, and $Requires does not guarantee a symbol

**MEASURED.** `display.create`, `display.setkeyenable`, `OBJ_TEXT` and `display.EVENT_*` each occur zero times in both reference manuals and zero times in the user manual. The API is specified only in `DMM6500DisplayDoc.md`, which Keithley distributes separately, and in `Display API Command Reference.md`, which ships at `TTI_Apps/TTI_Display_API/`. Version does not stand in for a feature test: `TTI_Apps/Pong.tspa` declares `-- $Requires: v1.7.0` and still reads `if display.setkeyenable then` at `:205` and `if display.EVENT_ENDAPP then` at `:222`. On 1.7.17a every symbol Pong guards is present — `display.create` and `display.setkeyenable` are functions, `display.EVENT_ENDAPP`, `display.EVENT_PRESS` and `display.OBJ_TEXT` all non-nil — so the guard costs nothing here and the feature test remains the only way to ask.

- `probe: print("REQ|ver="..tostring(localnode.version).."|create="..type(display.create).."|ske="..type(display.setkeyenable).."|endapp="..tostring(display.EVENT_ENDAPP).."|ep="..tostring(display.EVENT_PRESS).."|objtext="..tostring(display.OBJ_TEXT))` — the manual half stays a property of the manuals, settled by grep.

### Ten numbered fonts exist beside the four named ones, and display.FONT_11 is nil

**MEASURED.** Confirmed on firmware 1.7.17a. `FONT_SMALL`, `FONT_MEDIUM`, `FONT_LARGE` and `FONT_HUGE` exist too. Neither reference manual mentions any font constant — zero hits for `FONT_` in both.

- `probe: do local o="" for i=1,11 do o=o..i.."="..type(display["FONT_"..i])..", " end print(o, type(display.FONT_SMALL), type(display.FONT_MEDIUM), type(display.FONT_LARGE), type(display.FONT_HUGE)) end`

### display.create takes a numbered font although its own doc note lists only four

**MEASURED.** `Display API Command Reference.md:1446-1459` lists all fourteen under `display.setfont()`, marking `FONT_1` - `FONT_10` `(v1.7+)`; its `display.create()` note for `OBJ_TEXT` at `:1291-1294` lists four, and `DMM6500DisplayDoc.md:251` offers only "Large, Medium, or Small". Three files pass a numbered font straight to `display.create`: `TTI_Apps/CalculatorApp.tspa:198` and `TTI_Apps/DAQ6510_MultiFuncCtrl.tspa:493` use `FONT_7`, and `Series_2400_Graphical/SourceConstantPower24xx.tsp:119-129` uses `FONT_6` and `FONT_4`. The tree never calls `display.setfont` at all, and uses only three of the ten numbered sizes.

`display.create` accepts all ten: ten non-nil ids under one screen. `display.setfont` is a function too, and it takes a numbered font *and* a named one on a live object with nothing filed — but it does not erase what was there, so the old glyphs stay on the glass under the new ones until something repaints.

**The ten are not a size ladder.** Rendered as `"nWg"` and measured off the grab, heights run **15, 17, 20, 21, 26, 29, 32, 105, 21, 121 px**. `FONT_8` is 105 px, the same as `FONT_HUGE`; `FONT_9` drops back to 21 px and is the only *italic* face in the set; `FONT_10` at 121 px clips a four-character string. So a number chosen for size alone can silently change the typeface or overflow its object.

- `probe: do KTF={} KTS=display.create(display.ROOT,display.OBJ_SCREEN,"ktfonts") local x,y for i=1,10 do if i<=5 then x=15 y=15+(i-1)*90 else x=410 y=15+(i-6)*90 end KTF[i]=display.create(KTS,display.OBJ_TEXT,x,y,i.."Wg",0xFFFFFF,display["FONT_"..i]) end print("FONTS|scr="..tostring(KTS).."|f1="..tostring(KTF[1])) end` then `pcall(display.changescreen,KTS)` and grab the screen. A 90 px pitch runs 7 into 8 and 9 into 10; re-render those five at 165 px to measure them. Tear down every id and the screen; all are in named globals, so none can be lost.

### display.setkeyenable turns off a front-panel key, including TRIGGER

**MEASURED.** `TTI_Apps/Pong.tspa:206-209` disables `display.KEY_FUNCTION`, `display.KEY_HELP`, `display.KEY_TRIGGER` and `display.KEY_QUICKSET_APPS`. No reference manual or display document mentions `setkeyenable` or any `display.KEY_` constant. **The TRIGGER key is an escape route only while nothing has disabled it**, and `display.OFF` really does stop it posting `trigger.EVENT_DISPLAY`.

A model parked in `BLOCK_WAIT, trigger.EVENT_DISPLAY` with a one-reading measure block behind it, four arms, pressing the key the same way each time:

| arm | result |
|---|---|
| no press at all | `n = 0`, `STATE_WAITING` at block 1 |
| key enabled, pressed | `n = 1`, `STATE_IDLE` |
| key **disabled**, pressed | `n = 0`, `STATE_WAITING` |
| key enabled again, pressed | `n = 1`, `STATE_IDLE` |

Exactly those four `display.KEY_` constants exist, out of ten names asked: `KEY_HOME`, `KEY_MENU`, `KEY_EXIT`, `KEY_ENTER`, `KEY_APPS` and `KEY_QUICKSET` are all nil, so only Pong's set can be touched. **There is no getter**: `display.getkeyenable` is nil, and both calls are silent — `KEY_TRIGGER` to `display.OFF` and back to `display.ON` each returned nil with the event log at 0 after it. So a disabled key cannot be read back and **the socket cannot ask at all**: the crossover in the table above is reachable only through a press, by a finger or through the web panel. Nothing in this app calls `setkeyenable`.

- `probe: the press is injected through the instrument's own web front panel, not a finger — POST /ajax_proc with function=7 (BUTTON_PRESS_FUNCTION), button=8 (Trigger), session=<id>, then function=8 to release, read out of the instrument's /script/ajax_inc.js. The key-enable filter applies to that path, which is what makes the crossover evidence. Poll from INSIDE: do defbuffer1.clear() dmm.digitize.func=dmm.FUNC_DIGITIZE_VOLTAGE dmm.digitize.range=10 dmm.digitize.samplerate=1000 trigger.model.load("Empty") trigger.model.setblock(1,trigger.BLOCK_WAIT,trigger.EVENT_DISPLAY) trigger.model.setblock(2,trigger.BLOCK_MEASURE_DIGITIZE,defbuffer1,1) trigger.model.initiate() delay(4) local a,b,c=trigger.model.state() print("KP|n="..tostring(defbuffer1.n).."|st="..tostring(a)) trigger.model.abort() end`, with the press landing 1.5 s in. A HOST readback after the press measures nothing — the query aborts the model, see below.
- `probe: OFF and ON in ONE chunk, because a failure between them leaves the front panel's escape dead — do local r1,e1=pcall(display.setkeyenable,display.KEY_TRIGGER,display.OFF) local n1=eventlog.getcount(eventlog.SEV_ALL) local r2,e2=pcall(display.setkeyenable,display.KEY_TRIGGER,display.ON) local n2=eventlog.getcount(eventlog.SEV_ALL) print("SKE|off="..tostring(r1)..","..tostring(e1).."|n1="..tostring(n1).."|on="..tostring(r2)..","..tostring(e2).."|n2="..tostring(n2).."|getter="..type(display.getkeyenable)) end` — with no getter the counts are the whole verdict the socket has, and both are 0. The ten constants go one tagged read each.

### The display doc's two app-header rules are both wrong

**UNTESTABLE HERE** — the Apps menu is the only thing that parses these tags, so settling it needs a `.tspa` installed from the USB key and a human reading the menu. The evidence stands without a run: `DMM6500DisplayDoc.md:684,703` states two constraints, and the shipped apps break both unanimously.

| Rule in the doc | What the apps do |
|---|---|
| `appName` in `loadimage` "must match the name of the app used for the `$Title:` tag" | `loadimage`'s second argument is the `loadscript` identifier. `TTI_Apps/CalculatorApp.tspa` opens `loadscript CalcApp`, titles itself `TSP Calculator`, and closes `loadimage calc_icon CalcApp`. All eight icon-bearing apps do this. |
| A title "must not have spaces" | Seven of nine shipped apps use them, including `-- $Title: Res Tolerance`. |

- `no probe: a loadimage block INSTALLS an icon, and nothing in either manual or the display document removes one. On an instrument with a live app loaded that is an unbounded change to make for a readback, so it was left alone rather than run.`
- `no probe: which name the Apps menu binds the icon to, and whether a spaced title installs, need a USB install and a human reading the glass`

### A display escape code needs a trailing space only before a digit

**MEASURED.** Lua reads up to three decimal digits after a backslash, so only a following digit is ambiguous. `TTI_Apps/Probe_Hold.tspa:115` writes `"-\21-\21.\21-\21-\21-\21-\21"` and `TTI_Apps/Resistance_Tolerance_Meter.tspa:148` writes `string.format("%.5f\21\18", reading)`. `DMM6500DisplayDoc.md:628` says the codes "must have a space after them". On the instrument `"\21\18"` is **two** bytes, 21 and 18; `"\218"` is **one** byte, 218; `"\21 "` is two. So the doc's rule is right about the ambiguity and wrong to make it general — the space is needed only where a digit follows, and `Resistance_Tolerance_Meter` 's unspaced pair is correct as written.

- `probe: print("ESC|len2="..string.len("\21\18")..."|b1="..string.byte("\21\18",1)..."|b2="..string.byte("\21\18",2)..."|len1="..string.len("\218")..."|b218="..string.byte("\218",1)..."|lensp="..string.len("\21 "))`

### eventlog.suppress(N) appears in no manual, and what it suppresses is not the log entry

**MEASURED for the call, its scope and the log entry. What raises the dialog it is named for is unresolved, and the paragraphs below say so rather than pick a side.** `Application_Specific/Battery_Test/DC-IR/Battery_DCIR.tspa:614-617` calls it four times — 5076, 2732, 2731, 5078 — straight after `eventlog.clear()`. "suppress" has zero occurrences in both reference manuals and the user manual; the documented set is `clear`, `csv`, `getcount`, `next`, `post` and `save`. That app targets an SMU, and the DMM6500 takes it.

It suppresses the **dialog and not the entry** — suppressing 4919 and 4920 leaves both in the log. **What decides whether a 4915 has a dialog to suppress is unresolved, and it is none of the obvious candidates.** Seven cells from a socket file the entries and leave the panel clean — the first two rows are two cells each, run at `showevents` 0 and at 1:

| what posts the 4915 | buffer | rate | log | panel |
|---|---|---|---|---|
| `buffer.write.reading` ×20 past a capacity-10 `FILL_ONCE` buffer, at `showevents` 0 and 1 | scratch writable | no acquisition | ten 4915 | **clean** |
| a plain `BLOCK_MEASURE_DIGITIZE` storing 2,000 into 1,000, at `showevents` 0 and 1 | scratch standard | 40 kS/s | ten 4915 | **clean** |
| the same store, 25,000 into 20,000 | scratch standard | 1 MS/s | ten 4915 | **clean** |
| the **vendor repro**, model `STATE_WAITING` at block 4 with its 5 % reserve wrapping | scratch standard | 1 MS/s | ten 4915 | **clean**, native screen |
| the **vendor repro on** `defbuffer1`, the screenshot's own buffer | `defbuffer1` | 1 MS/s | ten 4915 and one **4918** | **clean**, app screen |

**The provocation path is not the discriminator.** A host `buffer.write` with no acquisition under it, a plain store overrunning its buffer, and an armed `LoopUntilEvent` whose pre-trigger reserve really was wrapping — `STATE_WAITING` at block 4, read from inside the chunk before the abort — all behave identically. Nor is it the **sample rate**, 40 kS/s and 1 MS/s giving the same thing; nor the **fill mode**; nor the **buffer**, since `defbuffer1` behaves exactly like a scratch one; nor the **active screen**, the app's and the native one both staying clean; nor `showevents` ; nor a stale suppression from another session — the last three rows ran from fresh sockets that never called `suppress` . The positive control holds in the same grab path: a bare `eventlog.suppress()` files **-109 Missing parameter** and its box, with **Details** and **OK**, is in the grab. Each grab follows that capture's own `abort()` by 0.5 s, which is the limit of the evidence — though the control's -109 box was still standing a second after its own call and through the grab that caught it.

**One difference is left, and it is not refuted.** `docs/vendor/img/4915-dialog.png` was raised from inside a **named script** — `t4915c` in the top bar, the panel in **WAIT** — where all seven cells are anonymous socket chunks; and the box carries "*Multiple errors have occurred*" below its buttons, so it may be a collapse box over a queue rather than a first-error box. That screenshot is this firmware and this event, so nothing here writes the dialog off. What is established is narrower and still worth having: **nothing reproducible from a socket raises it**, which is the only form an app can defend against anyway.

`docs/vendor/02-event-4915-unmutable.md` pins the mechanism for the **event**: while the model waits, the pre-trigger reserve keeps the most recent readings, so it wraps, and on a `FILL_ONCE` buffer every overwrite is a discard that posts the event. A plain measure block has no reserve to wrap and a `buffer.write` has no acquisition at all, and both still post ten. The same report measures the dialog asymmetry *across numbers*: `1130` and `2205` are error severity and raise no dialog, while a *warning*, `2874` , does — so the box belongs to the posting subsystem rather than to the severity or the number. Two boundaries go with it: the armed path produces **no** 4915 at all below somewhere between 10 and 50 kS/s, so a slow armed capture is silent for a different reason; and a clean panel is evidence only where the entries were filed, which all seven cells were — ten each — making them a negative and not a null.

**4918 exists.** The `defbuffer1` cell filed it once, between the tenth 4915 and `2728` . Its text was not captured, so the number is all that is known.

**It is scoped to the socket session, and there is no inverse.** `eventlog.unsuppress` is nil, as are `eventlog.count` and `eventlog.csv` . `suppress(4915,0)` , `suppress(4915,1)` and `suppress(0)` all return without raising and change nothing; a bare `suppress()` files **-109 Missing parameter** and raises its own dialog. A power cycle is not needed — a new connection has 4915 back.

**Where a dialog does appear, 4915 is only the first.** Suppressing it reveals **Warning 2728** "*Trigger model path #1 has been aborted*", filed by the model's own `abort()` . Suppressing 4915 and 2728 together leaves the panel clean through the whole provocation.

- `probe: do local a=pcall(eventlog.suppress,4915) local b=pcall(eventlog.suppress,2728) eventlog.clear() KB=buffer.make(20000,buffer.STYLE_STANDARD) KB.fillmode=buffer.FILL_ONCE dmm.digitize.func=dmm.FUNC_DIGITIZE_VOLTAGE dmm.digitize.range=10 dmm.digitize.count=20000 dmm.digitize.samplerate=1000000 trigger.model.load("LoopUntilEvent",trigger.EVENT_DISPLAY,5,trigger.CLEAR_ENTER,0,KB) trigger.model.initiate() delay(3) trigger.model.abort() delay(0.5) print("BOTH|a="..tostring(a).."|b="..tostring(b).."|ev="..tostring(eventlog.getcount())) end` — then grab the screen, because the panel is the verdict and the count is the same either way. The control is the same statement from a fresh socket with both pcalls removed.
- `probe: for the panel half, the vendor repro verbatim from a fresh socket with nothing suppressed — do KT=buffer.make(20000,buffer.STYLE_STANDARD) KT.fillmode=buffer.FILL_ONCE dmm.digitize.func=dmm.FUNC_DIGITIZE_VOLTAGE dmm.digitize.range=10 dmm.digitize.samplerate=1000000 dmm.digitize.count=20000 trigger.model.load("LoopUntilEvent",trigger.EVENT_DISPLAY,5,trigger.CLEAR_ENTER,0,KT) trigger.model.initiate() delay(3) local a,b,c=trigger.model.state() trigger.model.abort() print("V|st="..tostring(a).."|blk="..tostring(c).."|n="..tostring(KT.n)) end` — read the state from INSIDE the chunk and BEFORE the abort, or the premise goes unasserted: a model that never reached the wait at block 4 wraps no reserve, and a clean panel then means nothing. Repeat with `defbuffer1` in place of `KT` after `defbuffer1.fillmode = 0`, restoring it to 1. Grab the panel before any `display.changescreen`, which clears a dialog — see the next claim. Delete `KT` in a `pcall` and only nil it if the delete took, and put `samplerate` and `count` back.

### A modal event dialog is cleared by display.changescreen()

**MEASURED.** `DMM6500DisplayDoc.md` gives `changescreen` no effect on dialogs, and a box that stays until somebody presses a button would leave an unattended instrument unrecoverable. It does not: a bare `eventlog.suppress()` raises an **Error -109** box, and one `display.changescreen(scr)` onto the screen already in front takes it away. The log entry survives the dismissal — `eventlog.next` still returns -109 afterwards — so nothing is lost by clearing the glass.

- `probe: do pcall(eventlog.suppress) delay(1) print("RAISE|ev="..tostring(eventlog.getcount())) end` then `do local ok=pcall(display.changescreen,sdec.ui_scr) delay(0.5) print("CLEARDLG|ok="..tostring(ok)) end`, with a screen grab after each

### The instrument runs Lua 5.0.2, but its lexer takes | and 0b — two of these three idioms are fine

**REFUTED in part.** `_VERSION` reads `Lua 5.0.2`. Of the three idioms this entry called broken, only two are.

| Idiom | What happens |
|---|---|
| `math.fmod` | **Absent, as claimed.** `math.fmod` is nil and `math.mod` is a function. Twelve files use `math.mod`, including `TTI_Apps/DIOControlFull.tspa:29` and three DMM6500 streaming helpers. |
| `eventlog.SEV_WARN\|eventlog.SEV_ERROR`, and `digio.writeport(0b110101)` under "You can write binary, decimal or hexadecimal values" | **Both work.** `1\|2` evaluates to **3** and `0b110101` to **53** — Keithley's build extends the 5.0.2 lexer with a bitwise-or operator and a binary literal, so the manual is right and this entry was wrong. `bit.bitor(1,2)` is 3 and `bit.bitand(6,3)` is 2; `bit.bor` is nil. `eventlog.SEV_WARN` is 2. `TTI_Apps/Pong.tspa:518` keeps its mask line commented out for no reason that holds here. |
| `present_state == (trigger.STATE_WAITING or trigger.STATE_RUNNING)`, at `Amp-Hour_Measurement/Amp-Hours.tsp:80` | **Dead arm, as claimed.** `STATE_WAITING or STATE_RUNNING` yields `STATE_WAITING`, and `STATE_RUNNING == (STATE_WAITING or STATE_RUNNING)` is false. |

- `probe: compile it and then CALL it, because compiling is not evaluating — do local f=loadstring("return 1\|2") local g=loadstring("return 0b110101") local a,b,c,d if f~=nil then a,b=pcall(f) end if g~=nil then c,d=pcall(g) end print("LUA2\|orval="..tostring(b).."\|bval="..tostring(d).."\|bitor="..type(bit.bitor)) end` — `loadstring` returning a function was the first reading, and on its own it looks like a successful compile of something that cannot run.
- `probe: do local w,r=trigger.STATE_WAITING,trigger.STATE_RUNNING print("ORTRAP\|wor="..tostring(w or r).."\|eq="..tostring(r==(w or r))) end`

### The base library is Lua 5.0.2's, not the five functions the manual lists

**MEASURED.** The manual's "Base library functions" table holds `collectgarbage`, `gcinfo`, `tonumber`, `tostring` and `type`, and names `bit`, `io` and `opc` only among words you cannot use as variables. All twelve asked for are functions: `pcall`, `loadstring`, `unpack`, `pairs`, `ipairs`, `next`, `error`, `assert`, `gcinfo`, `dofile`, `opc` and `setmetatable`. `Application_Specific/Electrochemistry/EC_Framework.tsp` is built on `loadstring`, synthesising display code at run time in about twenty places (`:102`, `:148`, `:316`), and uses the tree's only `pcall` at `:402` to guard a compile. `unpack` appears in 21 files.

- `probe: print("BASE\|pcall="..type(pcall).."\|loadstring="..type(loadstring).."\|unpack="..type(unpack).."\|pairs="..type(pairs).."\|ipairs="..type(ipairs).."\|next="..type(next).."\|error="..type(error).."\|assert="..type(assert).."\|gcinfo="..type(gcinfo).."\|dofile="..type(dofile).."\|opc="..type(opc).."\|setmeta="..type(setmetatable))`

### Lua's io library is present, and it reports a failure that file.open cannot

**MEASURED.** `io` is a table with `open`, `close`, `read` and `write`, and a read-open of an absent file returns **nil plus `"/usb1/KTNOSUCH.XYZ: No such file or directory"`** — a named reason, where `file.open` gives no signal at all and files nothing. `io` handles are objects — `f:read()`, `f:write()`, `f:flush()` — not the integer descriptors `file.*` returns. `EC_Framework.tsp:389` reads `local f =`

`io.open('/usb1/'..filename, 'r')` then `f:read()`; `:442` and `:481` open `'w+'` and test `if f == nil then`; `Application_Specific/Ford EMC-CS-2009.1 specification/AWGforEMC.tsp:409` takes both returns as `file, err = io.open(filepath, "r")`. `io` appears in the reference manual only as a reserved word, and its 13 apparent `io.write` hits are all inside `digio.writeport`.

- `probe: do local f,e=io.open("/usb1/KTNOSUCH.XYZ","r") print("IOOPEN|f="..tostring(f).."|type="..type(f).."|err="..tostring(e)) if f then pcall(io.close,f) end end` — a read-open of an absent file creates nothing; the close is in a `pcall` because a record left open needs a power cycle.

### There is no 8.3 limit on /usb1 — a computed long name is written whole

**REFUTED.** By hand the vendor writes to the limit — `"/usb1/TestData.csv"`, `"/usb1/DuraQAAA.csv"`, `"/usb1/KI_setup.set"`. By program it writes past it: `Instrument_Examples/DMM6500/Save_Measurement.tsp:16-17` builds `file.open("/usb1/"..description.."csv", ...)` out of `display.input.string`, and `GAS/MS02_ExportData.tsp:22` uses `os.date("/usb1/data_%Y-%m-%d_%H-%M-%S.csv")`. The manual states no name limit **and there is none to state**: `file.open("/usb1/KTLONGNAME.CSV", file.MODE_WRITE)` returns a handle with zero events, and the file re-opens for read under **that full name** while the truncated `KTLONGNAM.CSV` does not exist. So the vendor's computed paths are safe and the hand-written 8.3 names are a habit, not a requirement.

- `probe: a write handle is not evidence — the file API never raises, so the name could have been truncated under it. The read-open is the proof: do local a=file.open("/usb1/KTLONGNAME.CSV",file.MODE_READ) local b=file.open("/usb1/KTLONGNAM.CSV",file.MODE_READ) print("EXISTS|long="..tostring(a).."|trunc="..tostring(b)) if a~=nil then pcall(file.close,a) end if b~=nil then pcall(file.close,b) end end`. `os.remove` is a function and removes it afterwards.

### os.time() returns fractional seconds, os.date() takes them, and os.clock() is the same wall clock

**REFUTED in part.** `os.time()` does carry a fraction — 0.811 of a second on one read — so `TTI_Apps/Pong.tspa:192-193` pacing an animation with `os.time() + 0.2` works. But `os.date("!%H:%M:%S", t)` **accepts the float** and prints the same string as the floored value, so `mainframeInfo.tsp:33-36`'s comment, "os.date() only accepts an integer timestamp", does not hold on a DMM6500. And `os.clock()` is **not a process clock**: it returns the same epoch value as `os.time()`, the two differing by 10 µs on one read. A difference across `delay(0.5)` still measures 0.50008 s, so `Speed_Scanning.tsp:9,59` bracketing a scan with it is sound — but the absolute value is wall time, not time since start. No manual has an *entry* for `os.time`, `os.date` or `os.clock`, but the reference manual uses both in examples — `os.time({year=...})` and `os.date('%c', gettime())` — so do not read this as absent, only as unspecified.

- `probe: do local a=os.time() local b=os.clock() local ok,d=pcall(os.date,"!%H:%M:%S",a) print("OSTIME|t="..tostring(a).."|frac="..tostring(a-math.floor(a))..."|clock="..tostring(b).."|dateok="..tostring(ok).."|date="..tostring(d)) end` — pass the raw float, not the floored one, or the claim's own question goes unasked.

### Nine introspection symbols appear in no DMM6500 manual because the DMM6500 does not have them

**REFUTED.** `Instrument_Examples/MP5000/mainframeInfo.tsp:46-58` prints `localnode.manufacturer`, `localnode.revision`, `localnode.description`, `localnode.autolinefreq`, `slot.autostart`, `gettimezone()`, `table.getn(userstring.table())`, `table.getn(script.table())` and `eventlog.count`. Sibling product, and **every one of them is nil here** — `script.table` and `userstring.table` are not functions but nil fields, so calling either raises *"attempt to call field `table' (a nil value)"*, and gettimezone() `raises *"attempt to call a nil value"*. Only` slot[1]` is a table. The manuals are silent because there is nothing to document: that script runs on an MP5000, not on this box.

- `probe: print("INTRO|mfr="..tostring(localnode.manufacturer).."|rev="..tostring(localnode.revision).."|desc="..tostring(localnode.description).."|alf="..tostring(localnode.autolinefreq).."|evc="..tostring(eventlog.count).."|gtz="..type(gettimezone).."|st="..type(script.table).."|ut="..type(userstring.table).."|sas="..tostring(slot[1]))` — `type()` first, then a guarded call, because a nil field and a function that fails read the same in prose and not in a probe.

### Keithley's own host code assumes five things the manual never states

**MEASURED,** for the three the instrument can answer. The reload collision is real: a second `loadscript` on a live name files **1408 "A script with the same name already exists"**, and `script.delete` itself logs **-104 "Data type error"** whichever way the script is named. The acknowledgement assumption holds too — `localnode.prompts` reads `localnode.DISABLE`, so the socket returns nothing until a script prints.

| Assumption | Evidence |
|---|---|
| Delete the script name before sending: `if loadfuncs ~= nil then script.delete('loadfuncs') end`. Twenty-six host files in Python, C#, VB and Go open with this line. | `Instrument_Examples/General/Send_a_Script_File_to_A_Keithley_Touchscreen_Instrument/..._Sockets_Python.py:225` |
| The 1024-byte ceiling is per line, not per write. The whole script goes in one `send`. | same file, `:228` |

| Assumption | Evidence |
| --- | --- |
| Nagle must be off: `setsockopt(IPPROTO_TCP, TCP_NODELAY, 1)`. | same file, `:51` |
| Nothing in the protocol acknowledges a command, so the acknowledgement is built out of `print`. Three of the four host-called functions end `print("ok")`; `get_data` answers with the data itself, and the file ends `print("functions loaded")` so the host can see the load finish. The host blocks on `recv`. | `00_Single_Channel/functions_V3.lua:39,55,61`; `Stream_TTI_SMU.py:102,108,147` |
| A query needs 100 ms between the write and the read. | `Drivers/DMM6500_DAQ6510/DMM6500_Python_Sockets_Driver/DMM6500_Sockets_Driver.py:100` |

- `probe: send loadscript ktst3 / print("a") / endscript TWICE with no delete between, then drain the log with eventlog.next(eventlog.SEV_ALL); clean up with do if ktst3~=nil then pcall(function() script.delete(ktst3) end) ktst3=nil end end and drain the −104 that leaves behind. print("HOST|prompts="..tostring(localnode.prompts).."|showevents="..tostring(localnode.showevents))` settles the acknowledgement.
- `no probe: TCP_NODELAY and the 100 ms sleep are host socket settings with no instrument readback. The per-line 1024-byte ceiling is assumed throughout this file rather than tested: every probe here is kept under it because past it the instrument answers −363 with no sentinel.`

### waitcomplete() holds the socket for the whole acquisition, which is what two Keithley scripts are warning about

**MEASURED, on the host clock.** `trigger.model.initiate() waitcomplete()` **in one chunk**, over a measure block of 80,000 readings at 40 kS/s, returned to the host in **2.002 s** against the capture's own 2.000 s, with `n = 80000` and `STATE_IDLE`; the same initiate with nothing behind it returned in **0.001 s**. So the host sees nothing for the length of the capture, and the vendor's imperative is sound: `Application_Specific/Battery_Test/DC-IR/Battery_DCIR.tspa:536` reads "or wait SRQ or *OPC if in python" and `Application_Specific/Rds(On)_of_SiC_MOSFET/Rdson_digitize_with_timing_marks.tsp:286` says "comment this out and wait SRQ if calling from Python". Neither says it stalls anything, both are SMU-paired application scripts, and the manual says only that the call "suspends the execution of commands", nothing about the remote interface. **The initiate and the wait have to go in the same chunk** — a host command aborts an initiated model, measured below, so an initiate the host then waits on separately has nothing left to wait for. The idle cost is small and separate: with nothing overlapped `waitcomplete()` returns in **15 µs** and `waitcomplete(0)` in **9 µs**, so the call is free when there is nothing to wait for and the stall belongs to the overlapped operation.

- `probe: do local t0=os.clock() waitcomplete() local t1=os.clock() local t2=os.clock() waitcomplete(0) local t3=os.clock() print("WC|idle="..tostring(t1-t0).."|grp0="..tostring(t3-t2)) end`
- `probe: the stall is timed on the HOST, because the host's wait is the claim. KR=buffer.make(300000,buffer.STYLE_STANDARD) with dmm.digitize.samplerate=40000, blocks 1 BUFFER_CLEAR KR and 2 MEASURE_DIGITIZE KR, 80000 − 2.000 s of capture by construction, and a capacity that cannot lap. Then time two sends: trigger.model.initiate() alone, and trigger.model.initiate() waitcomplete() print("WC2|n="..tostring(KR.n).."|st="..tostring(trigger.model.state())). n = 80000 with STATE_IDLE is what separates a call that waited from one that returned early. Abort between them, or the second initiate lands on a running model. Delete` KR `in a` pcall `on the way out and put` samplerate `and` count `back.`

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## The trigger model

### trigger.model.state() reports a lag, not a readiness — and it has nothing to do with the buffer

**REFUTED,** and this entry used to say the opposite. The time for an armed `LoopUntilEvent` model to report `trigger.STATE_WAITING` is **flat at about 0.19 s from 21,100 to 2,800,000 readings** — least-squares slope +1.6e-9 ± 5.7e-9 s per reading, t = 0.28, which is 4.5 ms across a 132× range. `digitize.count`, `samplerate` over a 1000× range, template `position`, a host `buf.clear()` and the poll cadence are all nulls. The earlier "about 1 ms at 21,100 readings" was never measured: thirteen arms at that size give **0.1668 to 0.2154 s**. What does move it is the **prior acquisition** — 0.228 s after aborting a model that was itself waiting, 0.191 s after `load("Empty")`, 0.177 s after a `SimpleLoop` run to idle, 0.164 s after a completed `digitize.read()`.

**The wait block is live about 2 ms after `initiate()`; the state call just does not admit it for another 190.** Two independent derivations agree: `pre-roll start = fire_s − N5/fs − n_pre/fs` gives 1.6–2.6 ms over six captures, and the block trajectory has `BUFFER_CLEAR` done at 0.0003 s and the zero `DELAY_CONSTANT` at 0.0012 s. Across twelve latency captures `STATE_WAITING` was **never once observed**, because the signal fired

inside one ramp period of the wait going live. So the state call is a lagging indicator and arming is not what costs the 190 ms. No vendor example loads `LoopUntilEvent`, and nothing in the tree or the manual states a block latency; `docs/TRIGGER.md:96` records that loading it destroys blocks 7–8, which bounds what "armed against a buffer" can mean.

- `probe: initiate a LoopUntilEvent model, poll trigger.model.state() from inside the instrument, and time the transition at two buffer sizes — but give the loop a delay. A spin that never yields starves the trigger engine: 56,102 state calls over 3 s and the model never left block 2, while 10, 1 and 0.2 ms cadences are indistinguishable. Full working in` /tmp/latency.md `and` /tmp/settle.md `.`

### A measuring model does NOT survive commands on the remote interface, and nor does a waiting one

**REFUTED.** Three host scripts query a meter in a loop for the whole length of an initiated model, and each depends on the model continuing — `get_data()` blocks inside the instrument until the buffer grows, so an abort on the first query would deadlock the host. `Streaming_Examples/00_Single_Channel/Stream_DMM6500.py:74,86` streams chunks; `01_Dual_Meter/KEIDMM6500_Stream_Measured_Dual_Meter.py:335,351` runs for 11 hours; `DAQ6510_Long_Term_Scan_with_Plotting_TSP.py:264,276` prints `defbuffer1.n` in a `while True`. **One of the three** ends with an explicit `trigger.model.abort()`; the other two let the host exit and leave the model running. **Only two are on a raw socket** (`Stream_DMM6500.py:105–106` and `DAQ6510_Long_Term…py:233`, both port 5025) and only one of those is a DMM6500 — the dual-meter script opens `USB0::0x05E6::0x6510::…::INSTR` under PyVISA, so it is USBTMC and one of its two boxes is a DAQ6510. One `print(defbuffer1.n)` from a socket aborts both shapes:

| model | first query | second query, 1 s later | log |
|---|---|---|---|
| inside `BLOCK_MEASURE_DIGITIZE`, 20,000 readings at 1 kS/s | `STATE_RUNNING`, n = 2 | **`STATE_ABORTED`**, n = 140 | `2731`, `2728` |
| parked in `BLOCK_WAIT, EVENT_ANALOGTRIGGER` that cannot fire | `STATE_RUNNING`, n = 12 | **`STATE_ABORTED`**, n = 200 | `2731`, `2728` |

**The block type is not the variable.** Both models above were digitizing when the abort landed, and `docs/TRIGGER.md:156` shows the same thing from the other direction — readings accumulating from 4,500 to 14,159 as each abort arrived. So "executing a measure block, never parked in a wait" bounded the wrong thing, and this entry used to say so. The first query is still answered truthfully, which is what makes it expensive to notice: the abort lands behind the reply and the model runs on about 140 ms. **The buffer is not one of the variables.** The same shape with a 200,000-reading measure block on a buffer made for the purpose was `STATE_ABORTED` with n frozen near 17,500 at the first read 1.5 s in, three runs over — and that run issued no query at all, which is the open mechanism `docs/TRIGGER.md` records. **The one shape measured to run to completion is one whose chunk never returned**: `initiate()` with `waitcomplete()` behind it in the same chunk took its full 80,000 readings to `STATE_IDLE`, so whatever ends a model early does not end one whose initiating chunk is still executing. **One candidate variable is still open** and worth a probe — a `printbuffer` read against a bare state poll. The residual conflict with the vendor is `Stream_DMM6500.py` alone, the one cited script that is both a DMM6500 and on a socket. The manual's "Aborting the trigger model" gives three stops and no command-interface effect.

- `probe: two steps, because the second command is the hypothesis. (a) one statement that arms and returns — do dmm.digitize.func=dmm.FUNC_DIGITIZE_VOLTAGE dmm.digitize.range=10 dmm.digitize.samplerate=1000 defbuffer1.clear() eventlog.clear() trigger.model.load("Empty") trigger.model.setblock(1,trigger.BLOCK_MEASURE_DIGITIZE,defbuffer1,20000) trigger.model.initiate() print("A_ARM|st="..tostring(trigger.model.state())) end. (b) from the host, print("A_POLL|n="..tostring(defbuffer1.n).."|st="..tostring(trigger.model.state())) three times a second apart, then drain. For the waiting case put a 200-reading measure block first — an EVENT_ANALOGTRIGGER wait cannot be block 1, the comparator only evaluates against a live stream — and set edge.level out of reach of the input, which a 2000-sample read measures rather than assumes (±10 mV idle here, so 9.9 V on the 10 V range).`

### Read a digitize block back: the constant is rewritten, and a template fixes both your count and your notify number

**MEASURED.** `trigger.BLOCK_DIGITIZE` exists and a block set with it **reads back as `MEASURE_DIGITIZE`**, confirming the rewrite. `trigger.COUNT_AUTO` is 2147483647, `COUNT_INFINITE` 2147483646 and `COUNT_STOP` 0; a hand-built `BLOCK_MEASURE_DIGITIZE` passed `COUNT_AUTO` took exactly the **7** readings `dmm.digitize.count` held. The notify collision is real too: a `LoopUntilEvent` template expands to six blocks ending `6) NOTIFY  NOTIFY: 2`, and `trigger.extout.stimulus` already defaults to `trigger.EVENT_NOTIFY2` — so a stimulus of ours on NOTIFY2 collides with a template's end-of-capture. `BLOCK_DIGITIZE` occurs zero times in the manual, and four examples set a block with it, two of them DMM6500: `Streaming_Examples/00_Single_Channel/functions_V3.lua:36`, `…/functions_V4_EXT_Trigger.lua:36`, `Amp-Hour_Measurement/Amp-Hours.tsp:61,63`. `Amp-Hours.tsp:61` digitizes without end by passing `trigger.COUNT_INFINITE`; `:63` is `COUNT_STOP`, commented "stop making digitized measurements". The manual admits one legacy spelling, `BLOCK_MEASURE`, and says the instrument rewrites it to `BLOCK_MEASURE_DIGITIZE`. **Both spellings are measured, and both are rewritten**: `setblock(1, trigger.BLOCK_MEASURE)` with no buffer and no count reads back as `1) MEASURE_DIGITIZE  BUFFER: defbuffer1 INITIAL MODE: DIGI INITIAL COUNT: 1`, nothing filed — so the firmware supplies the default buffer and the documented count of 1 as well as renaming the block. **Read a block back before trusting a name.** The **count** is not undocumented: a measure or digitize block with no count gets 1, stated with `setblock(1, trigger.BLOCK_MEASURE)` as the worked example, so only the spelling belongs here. For the count, `LIV/completePulse_CaseI_2601B_PULSE_DMM.tsp:288` and `LIV/measurementLIV_CaseI_2601B_PULSE_DMM.tsp:232` pass `trigger.COUNT_AUTO` after setting `dmm.digitize.count`; the manual gives that constant one phrase, "Use most recent count value", and never says which count it means. A template bakes a literal and takes it from the **buffer's capacity**, not from that setting: `load("LoopUntilEvent", trigger.EVENT_EXTERNAL, 50, trigger.CLEAR_ENTER, 0, defbuffer1)` against `defbuffer1`'s capacity of 100,000 expands to `3) MEASURE_DIGITIZE  INITIAL COUNT: INFINITE` and `5) MEASURE_DIGITIZE  INITIAL COUNT: 50000` while `dmm.digitize.count` stood at **7**. So `position` is a percentage of the buffer, the count is fixed at load time, and changing either the capacity or the count afterwards cannot move it — only a hand-built block reads the setting. The same split shows in the event numbering: `Grading_and_Binning_Resistors/Grading_and_Binning_Resistors.tsp:50` routes a GradeBinning template's end-of-test with `trigger.digout[6].stimulus = trigger.EVENT_NOTIFY2`, commented "Trigger pulse is output when the Notify Block generates an event" — the

same notify number the measured `LoopUntilEvent` expansion ends on, so a stimulus or a block of ours on `EVENT_NOTIFY2` collides with a template's end-of-capture. The GradeBinning page says EXTERNAL TRIGGER OUT is asserted at the end of each measurement and names no event.

- `probe: print(tostring(trigger.BLOCK_DIGITIZE)) dmm.digitize.func=dmm.FUNC_DIGITIZE_VOLTAGE trigger.model.load("Empty") trigger.model.setblock(1,trigger.BLOCK_DIGITIZE,defbuffer1,5) print(trigger.model.getblocklist()) print(eventlog.next(eventlog.SEV_ALL))` — a nil constant refutes it; a block list naming MEASURE_DIGITIZE confirms the rewrite.
- `probe: dmm.digitize.func=dmm.FUNC_DIGITIZE_VOLTAGE dmm.digitize.count=7 trigger.model.load("Empty") trigger.model.setblock(1,trigger.BLOCK_BUFFER_CLEAR,defbuffer1) trigger.model.setblock(2,trigger.BLOCK_MEASURE_DIGITIZE,defbuffer1,trigger.COUNT_AUTO) trigger.model.initiate() waitcomplete() print(defbuffer1.n, dmm.digitize.count) print(eventlog.next(eventlog.SEV_ALL))` — seven readings confirms it; no signal needed.
- `probe: trigger.model.load("Empty") trigger.model.load("LoopUntilEvent", trigger.EVENT_EXTERNAL, 50, trigger.CLEAR_ENTER, 0, defbuffer1) print(trigger.model.getblocklist()) print(defbuffer1.capacity, dmm.digitize.count) print(eventlog.next(eventlog.SEV_ALL)) trigger.model.load("Empty")` — six parameters, because four files 2747 and nothing arms. Read the notify number off the last block and the baked count off block 5, and read the capacity and `dmm.digitize.count` beside them: with the two set to the same number, which one the literal came from is unanswerable. No card and no signal, since LoopUntilEvent drives no digital line.

### A branch to block 0 ends the model

**MEASURED.** `LIV/completePulse_CaseI_2601B_PULSE_DMM.tsp:291` closes a model with `BLOCK_BRANCH_ALWAYS, 0` after a counter loop, and three sibling scripts do the same, five times over — all `Application_Specific/LIV_Diode_Testing/` scripts pairing a 2601B-PULSE with a "DMM7510 or DMM6510", building the blocks on `node[2]`, the DMM. The manual defines `branchToBlock` as "The block number to execute", gives 0 no meaning, and its own example branches from block 6 to block 20 in a six-block model. The instrument takes it: the block reads back as `3) BRANCH_ALWAYS  BRANCH_BLOCK: 0`, the model runs, stores its **one** reading and finishes `STATE_IDLE` with nothing worse than `2731` and `2732` in the log. **0 terminates.**

- `probe: do eventlog.clear() dmm.digitize.func=dmm.FUNC_DIGITIZE_VOLTAGE dmm.digitize.range=10 dmm.digitize.samplerate=1000 dmm.digitize.count=1 trigger.model.load("Empty") trigger.model.setblock(1,trigger.BLOCK_BUFFER_CLEAR,defbuffer1) trigger.model.setblock(2,trigger.BLOCK_MEASURE_DIGITIZE,defbuffer1,1) local ok=pcall(trigger.model.setblock,3,trigger.BLOCK_BRANCH_ALWAYS,0) print("ZERO|set="..tostring(ok)) end` — then `getblocklist()`, because a `setblock` that was refused returns true anyway, and only then initiate and read `defbuffer1.n` with `trigger.model.state()`.

### A wait on a line event arms with nothing acquiring, and one BNC releases it

**MEASURED** for the wait; the cable half is untested. A `BLOCK_WAIT, trigger.EVENT_EXTERNAL` as block 2, with the measure block behind it and nothing on the terminals, parks at **`STATE_WAITING`, block 2, `n = 0`** — so a line-event wait does arm with nothing acquiring, and the analog trigger's need for a live sample stream is the comparator's restriction and not the wait block's. `trigger.extin` and `trigger.extout` are both tables, `trigger.extout.assert` is a function, and with no card fitted `trigger.extout.stimulus` reads `trigger.EVENT_NOTIFY2` and `trigger.extin.edge` reads `trigger.EDGE_FALLING`. `LIV/completePulse_CaseI_2601B_PULSE_DMM.tsp:286` arms the meter on a hardware edge before any acquisition: block 1 clears the buffer, block 2 is `BLOCK_WAIT, EVENT_TSPLINK1, CLEAR_NEVER`, and the measure block is block 4. The omission is the point — the `BLOCK_WAIT` event table lists `EVENT_ANALOGTRIGGER` beside the line events with no distinction, so nothing marks the analog trigger as the one event that needs a live sample stream. That restriction belongs to the comparator, not to wait blocks. The same wait releases from the same box: `00_Single_Channel/functions_V4_EXT_Trigger.lua:14-15,30-37` sets `trigger.extout.stimulus = trigger.EVENT_NOTIFY1` with a block-1 NOTIFY, so EXTERNAL TRIGGER OUT fires when block 1 executes — and a BNC from OUT to IN lets the instrument satisfy its own `EVENT_EXTERNAL` wait, with `trigger.extout.assert()` releasing it from a socket.
`00_Single_Channel/functions_V3.lua:14-15,30-37` is the same shape over TSP-Link line 1 in `MODE_TRIGGER_OPEN_DRAIN`, needing no cable but the KTTI-TSP card. The manual is silent on a node waiting on a line it drives, and `trigger.extin` and `trigger.extout` carry no accessory-card note while every `digio.*` and `trigger.digin` / `digout` page does. Both probes substitute `EVENT_EXTERNAL` for the script's `EVENT_TSPLINK1` deliberately: the claim is about line events as a class, and the external one is the representative that needs no accessory card.

- `probe: do eventlog.clear() dmm.digitize.func=dmm.FUNC_DIGITIZE_VOLTAGE dmm.digitize.range=10 dmm.digitize.samplerate=1000 trigger.model.load("Empty") trigger.model.setblock(1,trigger.BLOCK_BUFFER_CLEAR,defbuffer1) trigger.model.setblock(2,trigger.BLOCK_WAIT,trigger.EVENT_EXTERNAL) trigger.model.setblock(3,trigger.BLOCK_MEASURE_DIGITIZE,defbuffer1,10) trigger.model.initiate() delay(0.5) local a,b,c=trigger.model.state() print("LINEWAIT|st="..tostring(a).."|blk="..tostring(c).."|n="..tostring(defbuffer1.n)) trigger.model.abort() end` — one statement, so no host command can abort it mid-wait.
- `no probe: the loopback needs one BNC from EXTERNAL TRIGGER OUT to EXTERNAL TRIGGER IN, and plugging it in costs a human at the bench. The readbacks above are the no-cable control, and they already show the wait parks.`

### A backward BRANCH_ALWAYS builds a model that cannot go idle

**MEASURED.** A count-1 measure block with `BLOCK_BRANCH_ALWAYS, 2` behind it acquired **500 readings in the first half-second and 1000 by the next**, held `STATE_RUNNING` throughout, and reached `STATE_ABORTED` only when `abort()` arrived — never `STATE_IDLE`. At 1 kS/s that is the full sample rate, so the loop costs nothing per lap; at 40 kS/s the same shape read **19,695 at 0.5 s and 39,395 at 1.0 s**, the full rate again. **And `abort()` does not stop it where it stands**: n read 39,395 before the abort and **39,400** after it, five readings later. That overshoot is what the 100 readings of capacity `sdec.acq_cap()` adds are there to absorb; with a buffer sized exactly to the count it would be a discard, and a discard is

another 4915. `01_Dual_Meter/functions_2021-03-06.lua:65` closes with `BLOCK_BRANCH_ALWAYS, 5` around a count-1 measure block. The model never reaches an end, `waitcomplete()` would never return, and the host's `abort()` is the only exit. `GAS/MS01_SetupAndRun.tsp:60,62` shows the escape its author abandoned, commented out: `BLOCK_BRANCH_ON_EVENT, trigger.EVENT_DISPLAY, 8` with `setblock(8, BLOCK_NOP)`, labelled "if the trigger key is pressed branch to where the trigger model is stopped".

- `probe: dmm.digitize.func=dmm.FUNC_DIGITIZE_VOLTAGE defbuffer1.capacity=10000 trigger.model.load("Empty") trigger.model.setblock(1,trigger.BLOCK_BUFFER_CLEAR,defbuffer1) trigger.model.setblock(2,trigger.BLOCK_MEASURE_DIGITIZE,defbuffer1,1) trigger.model.setblock(3,trigger.BLOCK_BRANCH_ALWAYS,2) trigger.model.initiate() delay(0.5) print(defbuffer1.n) print(trigger.model.state()) trigger.model.abort() delay(0.2) print(defbuffer1.n) print(trigger.model.state())` — n rising then frozen, and never IDLE until the abort. Do not call waitcomplete() on this model.
- `no probe: the abandoned TRIGGER-key escape needs a finger on the key, so EVENT_DISPLAY cannot be synthesised over a socket`

### Poll trigger.model.state() instead of blocking on waitcomplete()

**MEASURED.** One chunk, never from the host: mid-run the state is `STATE_WAITING` at block 2; `abort()` then `waitcomplete()` gives `STATE_ABORTED` at block 2; a re-`initiate()` returns to `STATE_WAITING` at block 2 and ran the measure block again, `defbuffer1.n` reaching 10 for two passes of a count-5 block. So **WAITING is a mid-run state, the abort/waitcomplete/re-initiate sequence works**, and the log carries `2731` / `2728` once per pass and nothing else. `TTI_Apps/KE_DAQ6510_Demo.tsp:35` is the clean form, `while present_state == trigger.STATE_WAITING or present_state == trigger.STATE_RUNNING do` — WAITING is a mid-run state, not a stall. `TTI_Apps/Resistance_Tolerance_Meter.tspa:243,247` takes all three returns and uses `block_num > 9` as a progress counter driven by the 0.25 s `display.OBJ_TIMER` at `:346`; at `:208-211` it re-arms with `abort()`, `waitcomplete()`, `initiate()`, though at `:137` and `:252` it aborts bare. Every trigger-model example in the reference manual blocks on `waitcomplete()`, and no text pairs `abort()` with `waitcomplete()` before a re-initiate.

- `probe: one chunk, never from the host — dmm.digitize.func=dmm.FUNC_DIGITIZE_VOLTAGE trigger.model.load("Empty") trigger.model.setblock(1,trigger.BLOCK_BUFFER_CLEAR,defbuffer1) trigger.model.setblock(2,trigger.BLOCK_MEASURE_DIGITIZE,defbuffer1,5) trigger.model.setblock(3,trigger.BLOCK_WAIT,trigger.EVENT_EXTERNAL) trigger.model.initiate() delay(0.4) local a,b,c=trigger.model.state() print(a,b,c) trigger.model.abort() waitcomplete() local d,e,f=trigger.model.state() print(d,e,f) trigger.model.initiate() delay(0.2) print(trigger.model.state()) trigger.model.abort()`

### A timer count is events per start, not events per model, and a timer can be aliased

**MEASURED.** Timer 2 with `count = 1`, started by `EVENT_TIMER1`, waited on once per lap of a three-lap `BLOCK_BRANCH_COUNTER`, produced **three readings** — so the count is per start, not per model. The alias works and so does writing through it while enabled: `local t = trigger.timer[1]` then `t.enable=1 t.delay=0.25 t.count=2` reads back as 0.25, 2 and `trigger.ON` on `trigger.timer[1]` itself, with **nothing filed**. It is the same object and not a copy — `t` and `trigger.timer[1]` print one address and compare equal — so there is nothing to synchronise and no stale copy to read. `Rds(On)_of_SiC_MOSFET/Rdson_digitize_with_timing_marks.tsp:110-111,126-127` chains timers and leaves the downstream counts at 1: timer 1 takes `count = NumPulses`, timer 2 takes `measDelay.count = 1` with `start.stimulus = trigger.EVENT_TIMER1`. The model waits on timer 2 once per lap of a `BLOCK_BRANCH_COUNTER` loop, so a count of 1 fires `NumPulses` times. The manual says to "make sure the count value is the same or more than any count values expected in the trigger model" and supplies the correct reading elsewhere: the count "sets the number of events to generate each time the timer generates a trigger event". The same script names its timers — `pulsePeriod = trigger.timer[1]` at `:90` — writes every setting through the alias, and sets `enable = 1` before `delay` and `count`, against the manual's advice to disable first.

- `probe: trigger.timer[1].reset() trigger.timer[2].reset() trigger.timer[1].delay=0.05 trigger.timer[1].count=3 trigger.timer[1].start.stimulus=trigger.EVENT_NOTIFY1 trigger.timer[1].start.generate=trigger.OFF trigger.timer[1].enable=1 trigger.timer[2].delay=0.01 trigger.timer[2].count=1 trigger.timer[2].start.stimulus=trigger.EVENT_TIMER1 trigger.timer[2].start.generate=trigger.OFF trigger.timer[2].enable=1 trigger.model.load("Empty") trigger.model.setblock(1,trigger.BLOCK_BUFFER_CLEAR,defbuffer1) trigger.model.setblock(2,trigger.BLOCK_NOTIFY,trigger.EVENT_NOTIFY1) trigger.model.setblock(3,trigger.BLOCK_WAIT,trigger.EVENT_TIMER2) trigger.model.setblock(4,trigger.BLOCK_MEASURE_DIGITIZE,defbuffer1,1) trigger.model.setblock(5,trigger.BLOCK_BRANCH_COUNTER,3,3) trigger.model.initiate() delay(2) print(defbuffer1.n, trigger.model.state()) trigger.model.abort() print(eventlog.next(eventlog.SEV_ALL))` — 896 bytes, so send it alone. Three readings from a count of 1 is the claim.
- `probe: local t=trigger.timer[1] t.enable=1 t.delay=0.25 t.count=2 t.start.generate=trigger.OFF print(trigger.timer[1].delay, trigger.timer[1].count, trigger.timer[1].enable, trigger.timer[1].start.generate) print(eventlog.next(eventlog.SEV_ALL))` — 0.25, 2 and enabled proves both the alias and the write-while-enabled.

### available(digio) works, and the manual's token list omits it

**MEASURED.** `available` is a function and **`available(digio)` is accepted**, returning `false` with nothing filed — so the token works and the manual's four-token list is incomplete. `digio`, `tsplink`, `channel`, `gpib` and `serial` are all tables here, so a nil namespace cannot be mistaken for a missing card: of the five only `available(channel)` is true, the other four false. `TTI_Apps/DIOControlFull.tspa:371` gates its whole digital-I/O setup on `if available(digio) == true then`. The `available()` entry lists four tokens — `channel`, `gpib`, `serial`, `tsplink` — while the Details on the same page name digital I/O among the accessory-card options this function exists "to check for".

- `probe: do print("AVAIL|digio="..type(digio).."|tsplink="..type(tsplink).."|channel="..type(channel)) local function t(x) local ok,v=pcall(available,x) if ok then return tostring(v) end return "ERR" end print("AVAIL2|digio="..t(digio).."|channel="..t(channel).."|ev="..tostring(eventlog.getcount())) end` — read `type()` first, because on a box with no card a nil token and a false answer are different findings, and drain afterwards: a refused token files an event rather than raising.

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## The digitizer and reading buffers

### The analog trigger level reads back changed, and it tracks the measured idle

**MEASURED.** On a 10 V digitize range, `dmm.digitize.analogtrigger.edge.level` written as 6.00 V reads back 5.99 V. Two runs apart the readback moved 20 mV, following the measured idle level. The level is adopted, not refused, at the top of the range. `dmm.digitize.range` on this app is 10, and the level range follows the active measurement range as documented. No example in the tree reads an `analogtrigger` attribute back, and the manual is silent on readback fidelity.

- `probe: write dmm.digitize.analogtrigger.edge.level = 6.0 on a 10 V range, read it back, change the applied idle level, read it back again`

### printbuffer takes a range the manual forbids

**MEASURED.** `printbuffer(1, defbuffer1.n, defbuffer1.readings)` over a five-reading buffer printed **exactly five real values and filed nothing**. **The manual forbids its own prescribed idiom.** Its `bufferVar.n` entry prescribes `printbuffer(1, bufferVar.n, bufferVar.readings)` as the way "to print all the readings in a buffer", and that form appears five times in the manual — while `printbuffer`'s own parameter table bounds `startIndex` by "more than one and less than endIndex" and `endIndex` by "less than the index of the last entry in the tables". Both bounds the table forbids are accepted, so the prescription is right and the table describes no behaviour this instrument has.

- `probe: do dmm.digitize.func=dmm.FUNC_DIGITIZE_VOLTAGE dmm.digitize.range=10 dmm.digitize.samplerate=1000 dmm.digitize.count=5 defbuffer1.clear() eventlog.clear() dmm.digitize.read() print("PB|n="..tostring(defbuffer1.n).."|ev="..tostring(eventlog.getcount())) end` then `printbuffer(1,defbuffer1.n,defbuffer1.readings)`, then drain. The clear matters: without it `defbuffer1` still holds the previous probe's readings and five values prove nothing.

### The LAN interface, not the digitizer, sets the streaming ceiling

**MEASURED,** for the response size and the digitizer's own ceiling. A `printbuffer(1,249,defbuffer1.readings)` under `format.REAL32` came back as **exactly 999 bytes in one chunk**, so the vendor's `chunkSize = 249` really is one Ethernet frame's worth and the arithmetic in this entry is right. The digitizer stops at **1,000,000 S/s**: asking for 2,000,000 or 10,000,000 leaves `dmm.digitize.samplerate` reading 1000000 and files `1130 "Parameter digitize sample rate, expected value from 1000 to 1000000"` — the request is refused, not clamped silently, and `format.SREAL` exists beside `REAL`, `REAL32`, `REAL64` and `DREAL`. `Streaming_Examples/00_Single_Channel/Stream_DMM6500.py:16` ships `sample_rate = 50000` and annotates it: "60kS/s is the max rate we have observed under certain conditions/circumstances. To attain higher sampling and data transfer rates, use USB." The same file caps a poll at `chunkSize = 249` `format.REAL32` readings — a `#0` header, four bytes a reading and a newline is 999 bytes, inside one Ethernet frame. 249 is tuning, not a ceiling: `01_Dual_Meter/KEIDMM6500_Stream_Measured_Dual_Meter.py:314,351` moves 1000 `REAL32` readings, about 4 kB, in one response and queries a second instrument for another — and being USBTMC it is weak evidence about a LAN ceiling. TCP segmentation hides the frame size, so the frame cannot be the cause. The manual ranks the interfaces — "USB has the fastest transfer rate" — and gives no figure and no response size.

- `probe: host counts the response bytes. dmm.digitize.count=300 dmm.digitize.read() format.data=format.REAL32 printbuffer(1,249,defbuffer1.readings) then restore format.data=format.ASCII` — expect exactly 999 bytes. For the two-column form, `dmm.digitize.count=1000 dmm.digitize.read() printbuffer(1,1000,defbuffer1.readings,defbuffer1.relativetimestamps)` — expect 8003 bytes and no truncation.
- `no probe: the sustained 60 kS/s figure needs a host-side timed extraction loop; from inside the instrument printbuffer blocks on socket backpressure, which bounds the rate but does not measure it`

### buffer.getstats(buf).n keeps counting after a buffer laps

**MEASURED.** A 100-reading `FILL_CONTINUOUS` buffer filled by a 500-reading measure block: `buf.n` pins at **100**, `startindex` 1, `endindex` 100 — and `buffer.getstats(buf).n` reads **500**. A second 500 into the same buffer takes it to **1000**, so it is cumulative since the buffer was made rather than per acquisition. It is also monotonic *during* a capture, which is the only property a poll loop needs: sampled a second apart inside one 30,000-reading block it read **10001, 20004, 30000** while `endindex` wrapped from 4 to 7. `buf.clear()` resets it to 0. So it is a strictly better progress signal than `endindex` for a lapping buffer, and the one thing to know is that `dmm.digitize.read()` asking for more readings than the destination's capacity stores **nothing at all** — `n = 0`, two `4928` entries — so the count has to come from a measure block, not a bare read. `.n` counts every reading ever stored, overwritten ones included, so it is the progress counter a lapping buffer still has. **The vendor leans on it across eight laps.** `01_Dual_Meter/functions_2021-03-06.lua:48` sets `samplebuff.fillmode = buffer.FILL_CONTINUOUS`, and its host runs **11 hours at 1 kS/s into a 5,000,000-reading buffer — about 39.6 M readings, eight times round** — polling `getstats().n` at `:94` throughout as the wait condition for the next chunk. Three more do the same with a private tally beside it: `00_Single_Channel/functions_V3.lua:69`, the line verbatim at `DMM7510/Stream_Digitized_Measurements/Sockets/Single_Box/functions_V3.lua:56`, and `Amp-Hours.tsp:81`. The first polls every 0.5 ms, so the call is cheap at size — though `EC_Framework.tsp:156-182` memoises `buffer.getstats` behind a hit-and-miss cache, so the vendor's own practice disagrees with itself on the cost. The manual gives `n` one line, "The number of data points on which the statistics are based".

- `probe: do KB=buffer.make(100,buffer.STYLE_STANDARD) KB.fillmode=1 eventlog.clear() dmm.digitize.func=dmm.FUNC_DIGITIZE_VOLTAGE dmm.digitize.range=10 dmm.digitize.samplerate=10000 trigger.model.load("Empty") trigger.model.setblock(1,trigger.BLOCK_MEASURE_DIGITIZE,KB,500) trigger.model.initiate() waitcomplete() local s=buffer.getstats(KB) print("GS1|n="..tostring(KB.n).."|si="..tostring(KB.startindex).."|ei="..tostring(KB.endindex).."|gsn="..tostring(s.n)) end` — `fillmode = 1`, numerically: `buffer.FILL_CONTINUOUS` does not exist on this firmware. Delete `KB` in a `pcall` and only nil it if the delete took.

### The real ceiling is 7,080,255 readings, across all buffers together

**MEASURED,** and the firmware states it itself. `buffer.make(9000000, buffer.STYLE_STANDARD)` files **`4920 "Invalid size: Reading buffer capacity must be between 10 and 7080255 readings"`** — so neither published figure is right, and the feature list's "seven million" is the closer of the two. The limit is on the **total**: with this app's 1,437,294-reading buffer and both default buffers at 100,000, a 6,000,000-reading request files `4919 "Available capacity for all reading buffers exceeded, remaining capacity is 5419050 readings"`, and 1,437,294 + 200,000 + 5,419,050 comes to 7,056,344. A second allocation reproduces that total from a different starting point: with the defbuffers again at 100,000, `buffer.make(3000000, buffer.STYLE_STANDARD)` is granted and reads capacity 3000000 back, and a further `buffer.make(4000000)` beside it files the same `4919` with `remaining capacity is 3857490 readings` — 3,000,000 + 200,000 + 3,857,490 = 7,057,490, within 1,146 readings of the first figure. **So a `4919` remaining-capacity figure is remaining beside whatever is already allocated, and reading it as the whole pool understates it by everything live at the time. A refused `buffer.make` returns nil while `pcall` reports success** — the only verdict is the readback, and deleting what it returned files `1138 "Parameter 1, expected type 'reading buffer', but found type 'unknown'"`. `bufferVar.capacity` says the overall capacity "can be up to 6,000,000 readings for standard reading buffers"; the feature list says "up to seven million". One example takes the feature list at its word and lands on the right number: `Use_the_DAQ6510_to_Control_a_Temperature_Chamber/F4T_EnvController_DAQ6510.tsp:521` sets `defbuffer1.capacity = 7000000`. `GAS/MS01_SetupAndRun.tsp:121` sidesteps both — asked for 10,000,000 by its own call site at `:212`, it guards on `if buffersize > 7000000` and sets `capacity = 0` to maximise. Capacity, not count, is the limit on a capture: `Amp-Hours.tsp:48` reads "Changing count is optional. The reading buffer capacity is the determining factor", beside `dmm.digitize.count = 1 -- CANNOT be zero; 1 to 55Million`.

- `probe: ask for MORE than the box can ever grant and read 4920's own text, which costs no allocation — do eventlog.clear() local ok,b=pcall(buffer.make,9000000,buffer.STYLE_STANDARD) if ok and type(b)=="table" then pcall(buffer.delete,b) end local e,m=eventlog.next(eventlog.SEV_ALL) print("POOL|ok="..tostring(ok).."|bt="..type(b).."|e="..tostring(e).."|m="..tostring(m)) end`. Do not walk upward through real sizes: a 4,500,000-reading `buffer.make` did not return inside 90 s and ended in `-286`, which looks exactly like a hung instrument.

### buffer.make on a name that already exists: the manual says overwrite

**MEASURED, and weighed against the pool.** The second `buffer.make` on the same global goes through silently: capacity 20 then capacity **30**, nothing filed. After one `buffer.delete` a fresh 20 can be made again, and the pool reads back unchanged afterwards — 100,000 on each default buffer and 1,437,294 on this app's — so the overwritten buffer is not stranded. Read off 4919's own remaining figure, the arithmetic closes exactly: against a baseline of **5,442,255** readings, one 100,000-reading buffer costs **102,375** of the pool, a **second** `buffer.make` on the same global costs **nothing more**, and one `buffer.delete` returns all 102,375 — net zero. So the first allocation is reclaimed as the overwrite happens, one delete frees everything the global ever held, and a buffer costs about 2.4 % more pool than the readings it holds. The manual documents the same-name case as "the existing buffer is overwritten by the new buffer", and TSP-side setup in the tree re-runs `buffer.make()` on the same global without deleting. The tree holds one `buffer.delete()` in total, at teardown of the 5,000,000-reading buffer in `KEIDMM6500_Stream_Measured_Dual_Meter.py:359`.

- `probe: do eventlog.clear() KTHA=buffer.make(20,buffer.STYLE_STANDARD) local c1=KTHA.capacity KTHA=buffer.make(30,buffer.STYLE_STANDARD) print("SAMENAME|cap1="..tostring(c1).."|cap2="..tostring(KTHA.capacity).."|ev="..tostring(eventlog.getcount())) end` — a distinctive global, because `buffer.make` might have refused a name already bound. Whether the first allocation came back is the 4919 readback, taken four times: baseline, after one 100,000-reading make, after a second make on the same global, and after one delete. Weigh it at 100,000 rather than at 20, so the delta is unmistakable, and take all four in one session: the baseline depends on what else is allocated and moves between them. Asking for 7,000,000 to read the figure allocates nothing, so the four questions are free.

### Indexing a buffer object returns the reading, writable styles included

**MEASURED.** Both forms agree, on both styles. A `STYLE_WRITABLE` buffer written with 1.25 reads `b[1]` **1.25** and `b.readings[1]` **1.25**, units `Volt DC`, nothing filed; on the digitizer-filled `defbuffer1`, `defbuffer1[1]` and `defbuffer1.readings[1]` both read -0.0092203052674 and `defbuffer1[defbuffer1.n]` gives the last reading. `TTI_Apps/Probe_Hold.tspa:227` indexes a `STYLE_WRITABLE` buffer as `App_buffer[App_buffer.n]`, `:275` reads `defbuffer1[endindex]`, and `Battery_DCIR.tspa:597` reads `dcirBuffer[j]`. The documented form is `bufferVar.readings[N]`, and `buffer.make()` warns that not all remote commands are "compatible" with the writable styles.

- `probe: do eventlog.clear() KB=buffer.make(10,buffer.STYLE_WRITABLE) buffer.write.format(KB,buffer.UNIT_VOLT,buffer.DIGITS_6_5) buffer.write.reading(KB,1.25) print("WIDX|idx="..tostring(KB[1])..."|sub="..tostring(KB.readings[1])..."|n="..tostring(KB.n).."|unit="..tostring(KB.units[1])) end`, then the same two forms on `defbuffer1`. Delete `KB` in a `pcall`, once.